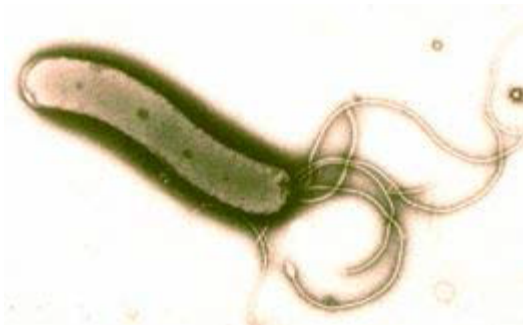
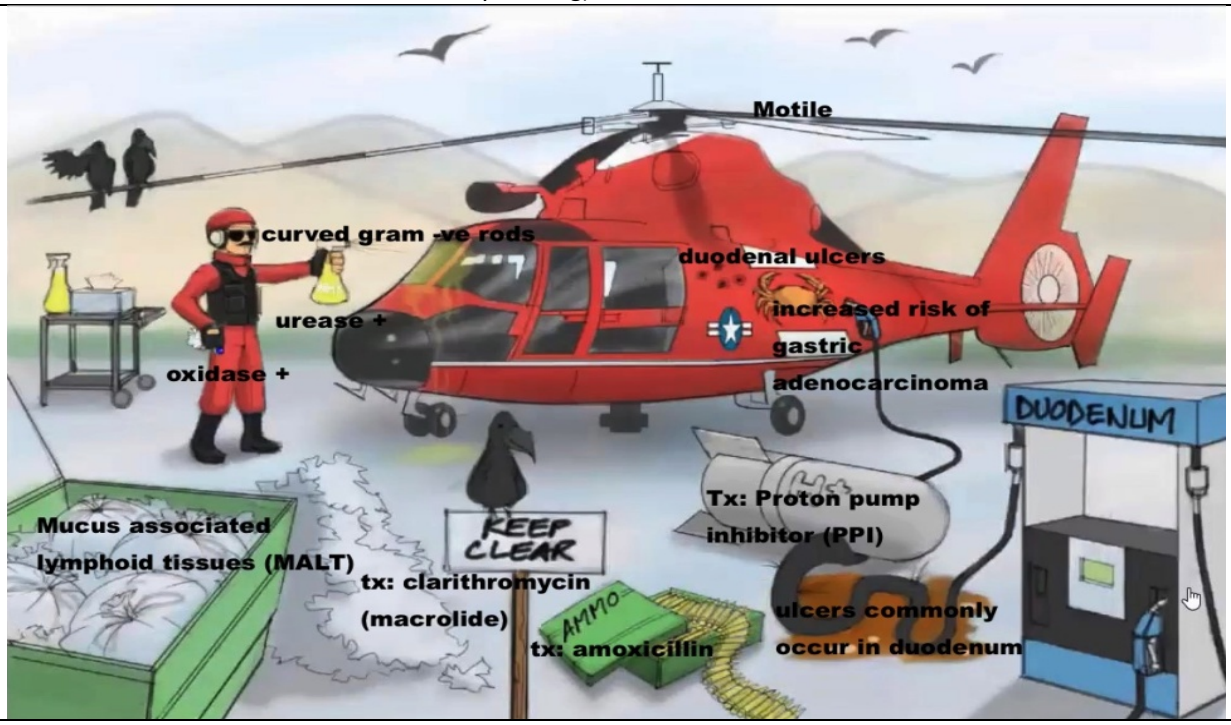
GENUS: ***Helicobacter***; SPECIES: ***pylori*** "Helicopter Pilot"



- ***General features:*** Gram-, spiral gastric rods, motile with flagella, curve shaped, microaerophilic, 370C growth, triple +: oxidase+, catalase+ and urease+
- ***Reservoir:*** humans, generally in the stomach near the pylorus
- ***Transmission:*** fecal-oral or oral-oral
- ***Pathogenesis:*** the urease creates an ammonium cloud that neutralizes the stomach acid, allowing its survival in the stomach acid during transit to the border (Colonizes mainly antrum of stomach); mucinase aids in penetration of mucous layer (rapid shift down to neutral as it penetrates). It invades into the stomach lining where the pH is neutral. Inflammation is prominent. There are two biotypes (I and II); type I produces vacuolating cytotoxin.
- ***Diseases:*** **chronic gastritis and duodenal ulcers (>95%)**
  - Associated with several forms of stomach cancer (gastric adenocarcinoma) that does not resolve with treatment of *H. pylori*; now classed by WHO as a type I carcinogen
  - Also associated with gastric mucosa-associated lymphoid tissue lymphoma (MALT-oma or B-cell lymphoma) that will resolve with treatment of *H. pylori*
- ***Diagnosis:***
  - Non-invasive screening test: Urease breath test: detect presence of urease: the test involves consuming a solution containing isotopically labeled urea which is degraded by urease into carbon dioxide and ammonia. The isotopically labeled carbon dioxide is absorbed into blood stream and exhaled in patient's breath (breath samples typically collected 30 minutes after labeled urea ingestion). The test has excellent sensitivity and specificity for both initial diagnosis of H.pylori and monitoring treatment success.
  - Invasive confirmatory test: Gastric biopsy with culture; histology with Giemsa or silver stain is gold standard;
  - Detection of H.pylori antigen in feces (fecal antigen test)
- ***Treatment:***
  - Treatment with triple therapy of **A**moxicillin (metronidazole if penicillin allergy) **+ C**larithromycin + Omeprazole, a **P**PI; treatment is prolonged for 10-14 days, up to 3 weeks. … **A**ntibiotics **C**ure ***P****ylori*.
  - Quadruple therapy by adding bismuth is used where clarithromycin resistance is >15% (PPI + bismuth + 2 antibiotics – metronidazole and tetracycle, eg)



Helicobacter Pylori: The helicopter Pilot

1. Red Helicopter -Curved Gram Negative rod
2. Mustache that is comma shaped - Helical slender curved rod Shape found in pylorus of the stomach
3. Not a rare infection.
4. Helicopters are motile - Motile by way of flagella
5. Ammonia bottle - Urease positive - MAJOR VIRULENCE FACTOR - allows reducing the acidity of the stomach and allows Helicobacter to survive there.
6. Can be tested with Urea breath test, radioactive urea that is broken down and exhaled as CO2 and NH3 by urease positive organisms.
7. Blue ring - Oxidase Positive – all curved rods are oxidase positive
8. Bullet holes in helicopter, Gas pump w/duodenum - Causes 95% of all duodenal ulcers
9. Crab on the helicopter - Mechanism of Chronic infection causes increased acid infection. At risk of developing gastric adenocarcinoma
10. Tissues that are thrown in garbage on the ground beside the boxes- Patient can develop lymphoma of mucous associated lymphoid tissue.
11. Treatment
    a. Gas Pump with duodenum and H+ Bomb - Proton Pump Inhibitor
    b. Amoxicillin - ammo
    c. Crow w/ Keep Clear - Macrolide - Clarithromycin
12. Transmission: a. Fecal Oral or Oral